Git, Generative AI, and Prompt Engineering

CERTIFICATE OF ACHIEVEMENT

This certificate is proudly presented to

Lesego Rabotapi

For: **Git, Generative AI, and Prompt Engineering**

With a score of: **46/50 (92%)**

Date: **October 24, 2025**

CAPACITI

Report Summary

Name	:	**Lesego Rabotapi**
Your Score	:	46 out of 50 (92%)
Correct Answers	:	46 Questions
Incorrect Answers	:	4 Questions
Unanswered	:	0 Questions
Required Passing Grade (%)	:	50%
Time Taken	:	22 mins 23 secs
Date	:	Oct 24, 2025
Your Result	:	Pass
Email	:	lesego.rabotapi@capaciti.org.za

Your Answers

Correct

Q1) Which process ensures code is reviewed before being integrated into main?

A. Creating a pull request (Your Answer)(Correct)

B. Running git rm

C. Adding tags

D. Switching repositories

Explanation:

Pull requests allow peers to review and approve changes before merge.

Correct

Q2) Your Generative AI model shows biased outputs. What should you inspect first?

A. Training-data diversity and quality (Your Answer)(Correct)

B. Git configuration

C. Merge workflow

D. Hardware speed

Explanation:

Bias often originates from imbalanced training datasets.

Correct

Q3) Which cloud platform provides tools designed to support building and deploying Generative AI applications?

A. Microsoft Azure

B. Google Cloud (Your Answer)(Correct)

C. Amazon S3

D. Oracle Cloud

Explanation:

Google Cloud offers AI-specific services for developing and hosting Gen AI apps.

Correct

Q4) As a project lead guiding a cross-functional AI team, what integrated practice most improves productivity and output quality?

A. Combining effective version-control workflows with clear AI prompt design (Your Answer)(Correct)

B. Removing remote repositories

C. Disabling commit tracking

D. Avoiding LLM integration

Explanation:

Strong workflow discipline and quality prompting boost collaboration and results.

Correct

Q5) You notice redundant commit messages in history. What is the best fix before merging?

A. Interactive rebase (Your Answer)(Correct)

B. Branch deletion

C. Force rewrite remote

D. Push new branch

Explanation:

Interactive rebase refines and cleans commit logs.

Correct

Q6) A developer wants an intuitive way to visualize commit history and branching. Which approach best supports this need?

A. Using graphical Git clients for visualization (Your Answer)(Correct)

B. Deleting local repository branches

C. Programming directly in the terminal

D. Using spreadsheets for tracking

Explanation:

GUI Git tools like Sourcetree provide visual branch and commit overviews.

Correct

Q7) Which prompting practice improves an AI model's accuracy?

A. Abstract inputs

B. Clear, structured, and context-rich instructions (Your Answer)(Correct)

C. Unrelated keywords

D. Minimal phrasing

Explanation:

Precision and context in prompts lead to more reliable outputs.

Incorrect

Q8) In Generative AI, which of the following is a subset of Deep Learning?

A. Supervised learning (Your Answer)(Incorrect)

B. Generative AI (Correct)

C. Reinforcement learning

D. Linear regression

Explanation:

Generative AI depends on deep-learning architectures to create new data.

Correct

Q9) Your AI model is producing irrelevant answers. What's the most probable issue?

A. Prompt lacks context (Your Answer)(Correct)

B. Insufficient GPU

C. Excess training data

D. Corrupt Git repo

Explanation:

Contextually weak prompts reduce response accuracy.

Correct

Q10) In Generative AI, 'hallucinations' refer to:

A. Visualization features

B. AI outputting incorrect information (Your Answer)(Correct)

C. Dataset noise

D. Hardware latency

Explanation:

Hallucinations are fabricated or irrelevant AI responses.

Incorrect

Q11) If a Git repository uses fetch, what occurs?

A. Commits are pushed to remote (Your Answer)(Incorrect)

B. Updates are retrieved without merge (Correct)

C. Branches are automatically merged

D. Local commits are deleted

Explanation:

Fetch downloads changes without altering local code.

Correct

Q12) Remote teamwork with Git and Bitbucket helps developers to:

A. Work in isolation

B. Push and sync branches collaboratively (Your Answer)(Correct)

C. Erase history

D. Skip commit messages

Explanation:

Git enables shared repositories for synchronized development.

Correct

Q13) In Generative AI, “Transformer” models are associated with:

A. Data encryption

B. Deep architectures for sequence processing (Your Answer)(Correct)

C. File compression

D. Statistical analysis only

Explanation:

Transformer architectures power LLMs like GPT for text generation.

Correct

Q14) What does the term prompt refer to in prompt-based AI systems?

A. A computer bug

B. A training dataset

C. An instruction or question to an AI model (Your Answer)(Correct)

D. A Git command

Explanation:

A prompt is a user input that directs AI output generation.

Correct

Q15) A prompt creates incomplete answers. How do you improve results?

A. Add explicit criteria and formatting (Your Answer)(Correct)

B. Remove all context

C. Randomize words

D. Shorten input arbitrarily

Explanation:

Detailed instructions yield full, structured responses.

Correct

Q16) Which of the following topics focuses on managing collaborative workflows and branching strategies in Git?

A. Cloud-infrastructure scaling

B. Git workflows and branching models (Your Answer)(Correct)

C. Neural-network optimization

D. Data visualization and querying

Explanation:

Git workflows define how teams branch, merge, and coordinate work.

Correct

Q17) Rebase is appropriate when you want to:

A. Rewrite commit history before sharing (Your Answer)(Correct)

B. Merge unrelated repositories

C. Revert deleted files

D. Change usernames

Explanation:

Rebasing creates a cleaner commit sequence pre-merge.

Correct

Q18) Who originally developed the distributed version control system widely used in software development today?

A. Larry Page

B. Andrew Ng

C. Linus Torvalds (Your Answer)(Correct)

D. Jeff Dean

Explanation:

Linus Torvalds built Git to manage Linux-kernel development efficiently.

Correct

Q19) To reduce AI model 'hallucinations,' you should:

A. Use vague prompts

B. Add contextual grounding (Your Answer)(Correct)

C. Increase temperature randomization

D. Ignore training data bias

Explanation:

Relevant context keeps responses factual and focused.

Correct

Q20) After resolving merge conflicts, you decide to squash commits before pushing. Why?

A. To simplify commit history (Your Answer)(Correct)

B. To erase change records

C. To lock branches

D. To alter remote IDs

Explanation:

Squashing combines commits for a cleaner timeline.

Correct

Q21) A developer building a chatbot using an LLM API needs to provide clear requests to the model. What principle is essential?

A. Ambiguous language

B. Structured and context-rich prompts (Your Answer)(Correct)

C. Random word patterns

D. Model-independent queries

Explanation:

Explicit, contextual prompts improve LLM response quality.

Correct

Q22) How can teams prevent unfinished features from affecting production code?

A. Work on feature branches (Your Answer)(Correct)

B. Commit to main directly

C. Avoid branching

D. Pull random changes

Explanation:

Feature branches isolate ongoing work safely.

Correct

Q23) When resolving merge conflicts, developers should:

A. Delete both branches

B. Choose any remote branch

C. Manually edit conflicting sections (Your Answer)(Correct)

D. Always prioritize remote changes

Explanation:

Manual editing is required to merge conflicting code safely.

Correct

Q24) If a developer rebases a branch others have already pulled, what risk appears?

A. Broken shared commit references (Your Answer)(Correct)

B. Merge auto-approval

C. Extra fast-forwarding

D. Branch duplication

Explanation:

Rebase rewrites IDs, causing desynchronization for teammates.

Correct

Q25) A system generates source code from text instructions. Which AI application does this demonstrate?

A. Generative AI for code creation (Your Answer)(Correct)

B. Git branching

C. Data labeling

D. Syntax parsing alone

Explanation:

Generative models can synthesize code based on natural language.

Correct

Q26) Why might a developer perform a rebase before merging changes?

A. To create a linear, readable history (Your Answer)(Correct)

B. To duplicate commits

C. To auto-merge others' branches

D. To reduce lines of code

Explanation:

Rebasing streamlines history for clarity and maintenance.

Correct

Q27) A commit in Git represents:

A. A programming language

B. A saved change to the repository (Your Answer)(Correct)

C. A copy of a branch

D. A bug report

Explanation:

A commit records a snapshot of repository changes.

Correct

Q28) A core difference between Generative AI and traditional machine learning is:

A. Generative AI creates new data (Your Answer)(Correct)

B. Generative AI stores static models

C. Traditional ML analyzes numerical data only

D. Generative AI cannot learn patterns

Explanation:

Generative AI produces new outputs similar to training data.

Correct

Q29) A business wants to automatically shorten detailed customer feedback into concise summaries for easier review. Which natural-language task is this describing?

A. Expanding

B. Transforming

C. Summarizing (Your Answer)(Correct)

D. Inferring

Explanation:

Summarization condenses lengthy text while preserving meaning.

Correct

Q30) What is Git primarily used for?

A. Managing spreadsheets

B. Version control and tracking changes in code (Your Answer)(Correct)

C. Running machine learning models

D. Data visualization

Explanation:

Git manages source-code versions and tracks project history.

Correct

Q31) Hotfixes must be released quickly without interrupting other work. What approach is best?

A. Gitflow hotfix branches (Your Answer)(Correct)

B. Forking workflow

C. Centralized only

D. No branching

Explanation:

Dedicated hotfix branches allow urgent patches independently.

Correct

Q32) If a feature branch is merged into main, what happens to its commits?

A. They disappear from history

B. They integrate into main's timeline (Your Answer)(Correct)

C. They overwrite main entirely

D. They become remote-only

Explanation:

Merging preserves commits within the main branch history.

Correct

Q33) What command brings commits from a remote repository into your local one?

A. git commit

B. git pull (Your Answer)(Correct)

C. git init

D. git branch

Explanation:

git pull fetches and merges remote changes locally.

Correct

Q34) What is the benefit of using a Gitflow-style workflow?

A. Automatic hardware upgrades

B. Structured branching for development and release (Your Answer)(Correct)

C. Reduced internet usage

D. Avoidance of branching

Explanation:

Gitflow defines consistent branch and release structures.

Correct

Q35) You're assigned to create an internal tool that condenses long reports into short summaries. Which AI concept and development process are most relevant?

A. Text summarization and prompt design (Your Answer)(Correct)

B. Branch protection and commit tagging

C. Remote workflow tracking

D. Data compression

Explanation:

Effective prompt structures enable accurate text summaries.

Correct

Q36) Which type of machine learning uses labeled examples for training?

A. Unsupervised learning

B. Generative AI

C. Supervised learning (Your Answer)(Correct)

D. Reinforcement learning

Explanation:

Supervised models learn from annotated input–output pairs.

Correct

Q37) An engineer prefers rebasing over merging on a private branch. Why?

A. To maintain linear project history (Your Answer)(Correct)

B. To hide conflicts

C. To duplicate code

D. To ignore remote changes

Explanation:

Rebase creates a simple, linear commit sequence for clarity.

Correct

Q38) Rewriting Git history should only be done when:

A. Commits are not shared (Your Answer)(Correct)

B. The branch is protected

C. The repository is public

D. You plan to erase all commits

Explanation:

Editing commit history is safe only before sharing with others.

Incorrect

Q39) Which of the following is a graphical Git client that provides an interface for managing repositories?

A. GitFlow

B. GitHub (Your Answer)(Incorrect)

C. Sourcetree (Correct)

D. Mercurial

Explanation:

Sourcetree is a GUI for performing Git operations visually.

Correct

Q40) Which cloud-based ecosystem is most suitable for training and deploying modern AI models, including Generative AI?

A. Google Cloud Platform (Your Answer)(Correct)

B. Dropbox

C. GitHub Pages

D. Microsoft Excel

Explanation:

GCP provides AI/ML infrastructure for model development and deployment.

Correct

Q41) If a team uses feature branches for each new functionality, which concept are they applying?

A. Rewriting history

B. Gitflow workflow (Your Answer)(Correct)

C. Centralized workflow

D. Patch management

Explanation:

Gitflow encourages developing features on isolated branches.

Correct

Q42) Bitbucket is primarily used for:

A. Hosting web applications

B. Storing movie files

C. Hosting Git repositories remotely (Your Answer)(Correct)

D. Building APIs

Explanation:

Bitbucket is a remote Git-repository hosting service.

Incorrect

Q43) A user wants to fix a typo in the most recent commit. Which action applies?

A. Cherry-pick

B. Rebase (Your Answer)(Incorrect)

C. Amend commit (Correct)

D. Merge fast-forward

Explanation:

git commit --amend updates the latest commit without adding a new one.

Correct

Q44) The git push command is used to:

A. Create a local repository

B. Send commits to a remote repository (Your Answer)(Correct)

C. Restore deleted files

D. Track local changes only

Explanation:

Push uploads local commits to remote storage.

Correct

Q45) You manage a project with multiple contributors. Which Git strategy ensures organized integration and review before merging?

A. Gitflow workflow with pull requests (Your Answer)(Correct)

B. Editing directly on main

C. Skipping branching

D. Emailing code files

Explanation:

Gitflow and pull requests enable structured collaboration.

Correct

Q46) Which of the following tasks falls under 'Transforming text'?

A. Grammar correction (Your Answer)(Correct)

B. Generating new stories

C. Code summarization

D. Sentiment classification

Explanation:

Transform tasks involve rewriting or translating existing text.

Correct

Q47) What is the primary purpose of learning prompt engineering when working with large language models (LLMs)?

A. To design and refine inputs that produce accurate and useful AI outputs (Your Answer)(Correct)

B. To improve internet search results

C. To calculate model accuracy

D. To monitor server uptime

Explanation:

Well-crafted prompts guide AI models to deliver precise and relevant responses.

Correct

Q48) You're building an email draft generator that deploys via Git. Which technologies work together for this pipeline?

A. Git and OpenAI API (Your Answer)(Correct)

B. Git and Photoshop

C. MySQL and Docker

D. Excel and Bash

Explanation:

Git handles versioning and deployment; OpenAI API generates content.

Correct

Q49) A fast-forward merge in Git occurs when:

A. No divergence exists between branches (Your Answer)(Correct)

B. A merge conflict appears

C. The network fails

D. The repository is corrupted

Explanation:

Fast-forward merges happen when branches share a linear history.

Correct

Q50) “Inferring” tasks in LLMs involve:

A. Predicting sentiment or topic (Your Answer)(Correct)

B. Creating stories

C. Translating languages

D. Compressing data

Explanation:

Inference extracts facts such as tone or subject from text.